

OMEYIMI BAYINAT MUSTAPHA

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PROFILE

Final-year Computer Engineering student at Covenant University, with hands-on experience in reinforcement learning for legged robotics, embedded systems, and control theory. Completed a curriculum reinforcement learning pipeline for quadruped terrain adaptation as a final-year research project, using MuJoCo and Python, achieving a sevenfold improvement in flood-terrain traversal over a scratch baseline. Proficient in ROS2, Python, Linux, and Git, and comfortable debugging across hardware and software stacks.

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, ROS2, MuJoCo, Git, Linux, Arduino

Machine Learning & Robotics: Proximal Policy Optimization (PPO), curriculum reinforcement learning, reward shaping, sim-to-real transfer, PID control, embedded systems integration

RESEARCH INTERESTS

- Terrain adaptation and gait generalization in legged robots using reinforcement learning, with a focus on real-world deployment.
- Physical AI safety, with a focus on understanding failure modes in embodied systems and building policies that hold up under unexpected conditions.
- Sim-to-real transfer and the role of classical control theory in guiding policy design.

RESEARCH

Final-year research project, Covenant University. Trained a Boston Dynamics Spot quadruped to walk and adapt across terrains using reinforcement learning in MuJoCo simulation.

- Designed a reward function incorporating forward velocity, gait phase incentives, and symmetry penalties to shape stable locomotion on flat terrain.
- Trained a PPO baseline policy that achieves a consistent diagonal trot gait on flat ground, reaching a forward velocity of around 0.6 m/s.
- Designed a curriculum learning pipeline that progressively exposed the agent to rough, flooded, and snow-covered terrain, producing a single generalized policy.
- Built MuJoCo XML scenes for each environment with unique friction values and terrain features, and benchmarked the flat-ground policy on each to establish a pre-training baseline.
- Achieved the strongest result on flooded terrain: the curriculum-trained policy averaged 122 m of traversal versus 17 m for a from-scratch baseline (roughly 7x), while the stall rate dropped from 30% to 10%, with zero falls recorded across all scenes.

WORK EXPERIENCE

Robotics & Control Intern

Mar 2025 – Oct 2025

Unicon Group, Abuja, Nigeria. Engineering internship focused on the design, testing, and deployment of robotic systems.

- Tested sensors, motors, and actuators to evaluate how they perform across different operating conditions.
- Wrote and debugged Arduino and Python scripts for robot simulations and experiments, logging results to support the team's engineering decisions.
- Tracked down and fixed integration issues between hardware and software on various robotic platforms.
- Set up and calibrated sensor modules, checking signal accuracy before physical deployment.
- Ran test cycles with the engineering team to validate control algorithms and record performance results.
- Kept documentation of test procedures and system setups up to date for the rest of the team.

Front-End Intern

Jul 2024 – Oct 2024

Heri Homes and Properties Ltd, Port Harcourt, Nigeria. Web development internship.

- Built and maintained production web pages using HTML, CSS, and Tailwind CSS, implementing responsive layouts from design mockups.
- Managed version control on GitHub through branching, code review, and pull request workflows in a collaborative team environment.

PROJECTS

Self-Balancing Robot

Embedded control project, real-time hardware and software integration.

- Built a two-wheeled balancing robot using a LilyGO ESP32 microcontroller and MPU6050 IMU, achieving stable hardware-level balance.
- Tuned a PID controller for real-time tilt correction using serial telemetry for live feedback during calibration.

AI Glasses for the Visually Impaired

SHENOVATION Hackathon, 2nd Place

Assistive technology prototype built under hackathon time constraints.

- Simulated a smart glass prototype in PyBullet incorporating obstacle detection and voice output for visually impaired users.
- Led a cross-functional team through design, build, and presentation, placing second in a nationally competitive hackathon.

Facial Door Recognition System

Embedded computer vision system for secure access control.

- Built a Raspberry Pi-based access control system integrating a Pi Camera, LCD display, and relay module to operate a physical lock.
- Implemented a face recognition pipeline covering enrolment, identity verification, and hardware actuation end to end.

Line-Following Robot

Foundational robotics project applying closed-loop control on embedded hardware.

- Designed and programmed an autonomous line-follower using IR sensors, motor drivers, and an Arduino microcontroller.
- Improved tracking reliability across varying track geometries through hands-on path-correction tuning and live debugging.

EDUCATION

Bachelor of Engineering, Computer Engineering, Covenant University, Ota, Nigeria

2021 – 2026 | CGPA: 4.20 / 5.00

Relevant Coursework: Control Systems, Applied Computer Programming, Digital Electronics, Artificial Intelligence and Machine Learning.

CERTIFICATIONS

Cisco IT Essentials, Cisco Networking Academy

Prompt Engineering, OBTranslate

Microsoft Word Specialist, Microsoft

LEADERSHIP & AWARDS

Robotics Lead, Robotics & AI Innovators Network, Covenant University	2025 – 2026
Robotics Lead, Google Developer Groups on Campus, Covenant University	2025 – 2026
Finalist, Pan African Robotics Competition	2024
2nd Place, SHENOVATION Hackathon, AI assistive device simulation	2025